International University of Business Agricultural and Technology

ASSIGNMENT ON STRUCTURED PROGRAMMING LAB

Emergency Potable Water Allocation for a Flood-Affected Municipality

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2. Problem Analysis

During a major flood, a municipality's temporary distribution centre has only 13,500 litres of potable water available, while the six affected service zones together request 20,400 litres. The problem is therefore a constrained-resource allocation problem: the requested quantity cannot be fully met, and a decision procedure is required to decide how much water each zone should receive.

The zones are not identical. Z01 (Municipal Hospital) and Z06 (Fire and Emergency Service) require a dependable minimum supply because lives may depend on it. Z02 and Z05 are emergency shelters housing displaced people who need immediate support. Z03 and Z04 are residential areas which, although less immediately critical, may have already gone without adequate water for one or more previous distribution cycles. Each zone also loses a different percentage of the water released to it (2%–12%) before it actually arrives, so the amount released from the centre is not the same as the amount a zone effectively receives.

Several requirements conflict directly with each other:

- Meeting every zone's minimum requirement may still leave hospitals and shelters short of a comfortable buffer, while ignoring minimums to fully serve one or two zones would leave others completely unserved.
- Serving zones strictly by “who asked for the most” or “who asked first” ignores urgency; serving equally regardless of need ignores the fact that hospitals must not run out.
- Favouring zones with a long previous waiting period is fair, but a newly declared emergency may need water immediately regardless of past waiting.
- Avoiding zones with a high distribution loss is efficient, but would unfairly and permanently disadvantage the population served by that zone.

Because of these conflicts, the assignment explicitly does not provide a fixed priority formula; the allocation strategy has to be designed, and it has to keep working correctly if the input values (available water, number of zones, or any zone's data) are changed — the program cannot simply hard-code the outcome for the sample dataset.

3. Proposed Solution

The allocation strategy adopted here has two independent stages: (1) a priority score decides the order in which zones are served, and (2) a two-phase release procedure decides how much water each zone actually gets, in that order, until the supply runs out.

3.1 Priority score

Every zone is given a numeric priority score built from two factors that the brief identifies as relevant to “nature of the service zone” and “previous waiting period”:

- Criticality weight, based on zone type: Hospital = 40, Emergency Service = 35, Emergency Shelter = 30, Residential = 20. This reflects that hospitals and emergency services need a dependable minimum supply, while shelters house displaced people who need immediate but slightly less time-critical support, and residential areas are the least immediately critical of the four categories.
- A waiting bonus of 5 points for every previous waiting cycle the zone has already endured. This lets a zone that has been repeatedly underserved rise in priority and, if it has waited long enough, even overtake a zone of a more critical type — directly addressing the “previous waiting vs. new emergency” conflict described in the brief.

Priority score = criticality weight + (5 × previous waiting cycles). This formula is intentionally simple and transparent so that its effect can be checked by hand, while still combining both required factors into a single comparable number.

3.2 Resolving equal priority

Where two or more zones end up with the same score, the assignment requires an explicit, justified tie-break rule rather than an arbitrary or hidden one. Three rules are applied in order, so the outcome is always deterministic:

- 1) The zone with more previous waiting cycles is served first (it has already waited longer for the same urgency level).
- 2) If still tied, the zone with the larger unresolved amount (requested − minimum) is served first, since it has more outstanding need beyond the guaranteed minimum.
- 3) If still tied after both checks, the zone that was entered first in the input keeps the earlier position. This final rule only matters when the first two genuinely cannot distinguish the zones, and it guarantees the programme always produces one, repeatable order rather than an undefined one.

3.3 Two-phase release procedure

Distribution loss means that the quantity released from the centre and the quantity a zone effectively receives are different: to guarantee a zone receives X litres, the centre must release X ÷ (1 − loss%). Because the assignment specifically states that “the minimum requirement should be considered in relation to the quantity effectively received”, every allocation decision in this programme is made in terms of the loss-adjusted release amount, not the raw requested/minimum figures.

- Phase 1 (minimum guarantee): in priority order, each zone is released enough water to deliver its own minimum requirement after loss, as long as supply remains. A zone that runs the centre out of water receives whatever is left as a partial allocation.
- Phase 2 (extra allocation): still in the same priority order, any zone whose minimum was fully met in Phase 1 is then topped up towards its full requested quantity (again loss-adjusted), until either it is fully served or the remaining supply is exhausted.

This two-phase design was chosen over a single pass because it guarantees that no zone receives “extra” water while a higher-priority zone remains below its minimum, which would contradict the requirement that hospitals and emergency services receive a dependable minimum supply first.

3.4 Should distribution loss affect priority?

A deliberate design decision was made not to let distribution loss influence the priority score itself, only the quantity that must be released. If a high-loss zone were also given a lower priority purely because it is “inefficient” to supply, a genuinely critical zone (for example a hospital with old, leaking pipework) would be penalised twice — once by needing more water, and again by being served later. The brief explicitly warns against this exact trade-off (“avoiding such a zone merely because of the loss may unfairly disadvantage the affected population”), so efficiency is handled only at the allocation-quantity stage, while fairness and urgency remain the only inputs to priority.

4. Programme Design

4.1 Data organisation

All information belonging to one service zone is kept in a single C structure (Zone), and all zones are stored in one array of structures (Zone zones[MAX_ZONES]). This was chosen instead of several separate parallel arrays (one array for requested amounts, one for minimums, etc.) because a structure keeps every value that belongs to the same zone physically together in memory. This directly satisfies the assignment's data-organisation requirements: any zone can be accessed by its array index or located by its Zone ID (searching), every zone can be visited systematically with a single for-loop (processing), calculated fields such as priority score, allocated water and condition are written back into the same structure instead of being kept in a separate array (updating), and the relationship between a zone's requested amount, its minimum, its loss and its own calculated results can never become mismatched (association).

4.2 Programme structure (functions)

The programme is divided into small, single-purpose functions:

- inputAvailableWater() / inputZones() — read the available supply and each zone's data; the number of zones is itself an input, so the programme is not limited to six zones.
- criticalityOf() / computePriority() — derive the priority score described in Section 3.1 for every zone.
- sortByPriority() — arranges the zones array into priority order using a selection sort together with the tie-break rule from Section 3.2.
- allocateWater() — performs the two-phase release procedure from Section 3.3 and returns the water left at the centre.
- evaluateDelivery() — converts each zone's allocated (released) amount into an effective delivered amount and a remaining shortage, using the loss percentage.
- classifyCondition() — labels each zone as Full Requirement / Minimum Satisfied / Below Minimum / Unserved based on the effective delivered amount.
- searchZoneByID() — a linear search that locates a single zone's full record by its Zone ID.
- indexOfHighestShortage() — a linear scan that identifies the zone with the largest unresolved requirement.
- printBanner() / printAllocationResult() / printPriorityOrder() / printAllocationSummary() — display the case banner, the priority-ordered allocation table, the explicit priority ranking, and the final summary (totals, counts of each service condition, and a check that total allocation never exceeds the available supply).

4.3 Algorithm (pseudocode)

```
READ availableWater, numberOfZones
FOR each zone: READ id, name, type, requested, minimum, loss%, prevWaiting

FOR each zone:
    weight = criticalityWeight(type)
    priorityScore = weight + 5 * prevWaiting

SORT zones DESCENDING by priorityScore,
     tie-break -> prevWaiting, then (requested-minimum), then input order

// Phase 1: guarantee minimum (loss-adjusted) in priority order
FOR each zone IN priority order WHILE water remains:
    releaseForMin = minimum / (1 - loss%)
    allocate min(releaseForMin, waterRemaining) to zone
    reduce waterRemaining accordingly

// Phase 2: top up toward full request (loss-adjusted), same order
FOR each zone IN priority order WHILE water remains:
    IF zone's minimum was not fully released, SKIP
    releaseForFull = requested / (1 - loss%)
    extra = releaseForFull - alreadyAllocated
    allocate min(extra, waterRemaining) to zone
    reduce waterRemaining accordingly

FOR each zone:
    effectiveDelivered = allocated * (1 - loss%)
    shortage = max(0, requested - effectiveDelivered)
    classify condition from effectiveDelivered vs requested/minimum

PRINT priority-ordered table and allocation summary
FIND and PRINT zone with highest shortage
SEARCH (optional) a zone by ID and print its full record
```

5. Optimisation Considerations

The programme visits the zone array a small, fixed number of times per run (once to read input, once to compute priority, once — inside a selection sort — to order the zones, twice inside allocateWater() for the two phases, once to evaluate delivery, once to classify condition, and once to print), so the overall cost grows only with the square of the number of zones (from the sort) and is otherwise linear. For the scale of this problem (a handful of service zones per distribution centre) this is efficient enough, and a more complex sorting algorithm was judged unnecessary and would have reduced readability without a meaningful performance benefit.

Every value that is calculated once — the priority score, the allocated amount, the effective delivered amount and the shortage — is stored back into the same Zone structure instead of being recomputed later when it is needed again (for example, when printing the summary or searching for the highest shortage). This avoids repeated, redundant calculation and keeps a single authoritative copy of each result, satisfying the requirement to avoid unnecessary duplication of data.

Each responsibility (input, priority, sorting, allocation, loss evaluation, classification, searching, printing) is isolated in its own function. This function decomposition keeps each piece of logic short enough to verify independently and means that, for example, the tie-break rule or the criticality weights can be changed in one place without touching the allocation or printing logic.

Memory use is a single fixed-size array of structures (no dynamic allocation, no duplicate arrays holding the same values in a different form), which is appropriate for the small, bounded number of zones expected at one distribution centre.

6. Implementation and Results

The solution is implemented as a single C source file, water_allocation.c, using only the standard library (stdio.h, string.h). The major techniques used are: an array of structures for data organisation; user-defined functions for every stage of processing; a selection sort with a multi-level tie-break comparison for ordering; a two-pass greedy loop for allocation; and linear search for looking up a zone by ID and for finding the zone with the greatest shortage. No zone name, zone ID or numeric value from the sample dataset appears anywhere in the allocation logic itself — all of it is read as input — so the same code produces correct results for the original dataset and for every modified dataset used in testing (Section 7).

6.1 Original dataset

Input: 13,500 litres available, 6 service zones (see dataset table below). Total requested = 20,400 L; total minimum requirement = 12,400 L, so the centre has enough to guarantee every zone's minimum but not enough to fill every request.

Zone ID	Service Zone	Type	Requested (L)	Min Req (L)	Loss (%)	Prev. Waiting
Z01	Municipal Hospital	Hospital	4000	3000	2	0
Z02	Central Flood Shelter	Emergency Shelter	3500	2500	4	1
Z03	Ward 3 Residential Area	Residential	4500	2000	8	2
Z04	Ward 5 Residential Area	Residential	3800	1800	12	3
Z05	School Emergency Shelter	Emergency Shelter	2600	1600	5	1
Z06	Fire and Emergency Service	Emergency Service	2000	1500	1	0

Actual console output produced by water_allocation.c for the original dataset (Case 1):

```
========== CASE 1 ==========
Original Dataset
Available Water: 13,500 L


==========================================================================
                                 ALLOCATION RESULT
```

```
================================================================================
ID   Zone                     Request   Minimum   Priority  Allocated  Delivered  Shortage  Status
--------------------------------------------------------------------------------------------
Z01  Municipal Hospital       4000      3000      40.00     3477       3408       592       MINIMUM
Z04  Ward 5 Residential Area  3800      1800      35.00     2045       1800       2000      MINIMUM
Z02  Central Flood Shelter    3500      2500      35.00     2604       2500       1000      MINIMUM
Z05  School Emergency Shelter 2600      1600      35.00     1684       1600       1000      MINIMUM
Z06  Fire and Emergency Service 2000      1500      35.00     1515       1500       500       MINIMUM
Z03  Ward 3 Residential Area  4500      2000      30.00     2174       2000       2500      MINIMUM
================================================================================

Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

Summary
------------------------------------------
Total Requested Water          : 20400.00 L
Total Allocated Water          : 13500.00 L
Total Effective Water Delivered: 12807.56 L
Total Unresolved Requirement   : 7592.44 L
Remaining Water                : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 6 zone(s)
Below Minimum                : 0 zone(s)
Completely Unserved          : 0 zone(s)

Check: total allocated (13500.00 L) does not exceed available supply (13500.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 2500.00 L
```

All six zones reached “MINIMUM” status but none reached “FULL”, and the programme correctly identified Z03 (Ward 3 Residential Area) as the zone with the highest unresolved requirement (2,500 L short), consistent with it having the lowest priority score in this dataset. The full allocated quantity (13,500 L) was released and 0 L remained at the centre.

7. Testing and Validation

The programme was tested under the six required cases plus one additional case designed independently, by editing only the input values that are typed in when the programme runs — the source code (water_allocation.c) was never changed between tests. The table below summarises the test conditions, and the actual console output captured from each run is reproduced immediately below it for verification.

#	Test Condition	Expected Result	Actual Result	Pass/Fail
1	Original dataset, 13,500 L available (base case).	Supply (13,500 L) is less than total request (20,400 L) but more than combined minimum (12,400 L). All 6 zones should at least receive their minimum; higher-priority zones may receive extra towards their full request; all released water is used up.	All 6 zones reached “Minimum Satisfied”. Total allocated = 13,500.0 L (all supply used). Remaining at centre = 0.0 L. Highest unresolved zone: Z03 (2,500 L short).	Pass
2	Available water raised to 22,000 L so it exceeds the loss-adjusted amount needed for every zone’s full request (≈ 21,694 L).	Every zone reaches “Full Requirement”; total effective delivered = total requested (20,400 L); some water remains unused at the centre.	All 6 zones = Full Requirement. Effective delivered total = 20,400.0 L. Remaining at centre = 306.0 L.	Pass
3	Available water reduced to 10,000 L, below the combined minimum requirement of	Not every zone can even receive its minimum. Lower-priority zones should end up Below Minimum or Unserved, while the	4 zones Minimum Satisfied, Z06 Below Minimum, Z03 (lowest priority) completely Unserved. Total allocated exactly 10,000.0	Pass

	12,400 L.	programme must not crash or allocate more water than is available.	L.	
4	Two new zones ("Shelter Alpha", "Shelter Beta") given identical type, requested, minimum and waiting-cycle values, to force an exact priority-score tie.	Both zones receive the same priority score; the tie-break rule (previous waiting → unresolved demand → input order) should place Alpha before Beta since Alpha was entered first, and both should be treated identically otherwise.	Alpha ranked immediately above Beta with identical allocation and condition values for both; order matched the input order as designed.	Pass
5	Distribution loss of Ward 5 Residential Area (Z04) increased from 12% to 60%, all else unchanged.	Z04 will need to have far more water released to still deliver the same 1,800 L minimum, consuming a larger share of the centre's supply and leaving less for the lowest-priority zone.	Z04's allocation rose from 2,045.5 L to 4,500.0 L for the same 1,800 L delivered; Z03 (lowest priority) dropped from Minimum Satisfied to Below Minimum.	Pass
6	Previous waiting cycles of Ward 3 Residential Area (Z03) increased from 2 to 8, all else unchanged.	Z03's priority score should rise enough to overtake every other zone, moving it from the lowest priority rank to the highest.	Z03 moved from priority rank 6 to priority rank 1 and became the first zone served.	Pass
7 (student-designed)	Available water set to exactly 12,400 L — numerically equal to the sum of the six raw minimum requirements, but before accounting for distribution loss.	Because meeting each zone's minimum requires releasing minimum ÷ (1 − loss), the true loss-adjusted requirement is higher than the raw sum of minimums. The lowest-priority zone (Z03) should therefore still fall Below Minimum, demonstrating that distribution loss silently reduces the effective capacity of the supply even when raw totals look sufficient.	5 zones Minimum Satisfied, Z03 (rank 6) Below Minimum with only 1,370.6 L effectively delivered against a 2,000 L minimum.	Pass

In every case, the programme printed an explicit check confirming that the total water allocated never exceeded the available supply, which the assignment specifically requires. Together, the seven cases exercise all five situations the programme is required to handle correctly: sufficient supply (Case 2), insufficient supply (Case 1), insufficient even for the combined minimum (Cases 3 and 7), equal priority (Case 4), and the supply becoming exhausted partway through allocation (Cases 1, 3, 5 and 7).

7.1 Case 1 — Original dataset (base case)

Expected: supply (13,500 L) is less than the total request (20,400 L) but more than the combined minimum (12,400 L), so every zone should at least reach its minimum, higher-priority zones may receive extra, and all supply should be used.

```
========== CASE 1 ==========
Original Dataset
Available Water: 13,500 L


================================================================================
                                ALLOCATION RESULT
================================================================================
ID   Zone                    Request   Minimum   Priority  Allocated  Delivered  Shortage  Status
--------------------------------------------------------------------------------------------
Z01  Municipal Hospital      4000      3000      40.00     3477       3408       592       MINIMUM
Z04  Ward 5 Residential Area  3800      1800      35.00     2045       1800       2000      MINIMUM
Z02  Central Flood Shelter   3500      2500      35.00     2604       2500       1000      MINIMUM
Z05  School Emergency Shelter 2600      1600      35.00     1684       1600       1000      MINIMUM
Z06  Fire and Emergency Service 2000      1500      35.00     1515       1500       500       MINIMUM
Z03  Ward 3 Residential Area  4500      2000      30.00     2174       2000       2500      MINIMUM
================================================================================


Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)
```

```
Summary
------------------------------------------
Total Requested Water          : 20400.00 L
Total Allocated Water          : 13500.00 L
Total Effective Water Delivered: 12807.56 L
Total Unresolved Requirement   : 7592.44 L
Remaining Water                : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 6 zone(s)
Below Minimum                : 0 zone(s)
Completely Unserved          : 0 zone(s)

Check: total allocated (13500.00 L) does not exceed available supply (13500.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 2500.00 L
```

Result: PASS — all 6 zones reached MINIMUM, all 13,500 L was allocated, 0 L remained, and Z03 was correctly flagged as having the highest shortage.

7.2 Case 2 — Supply sufficient for every full request

Expected: with 22,000 L available (more than the ≈ 21,694 L needed to satisfy every request after loss), every zone should reach FULL status and some water should remain unused.

```
========== CASE 2 ==========
Sufficient Water - All Zones Receive Full Requested Quantity
Available Water: 22,000 L

================================================================================
                                ALLOCATION RESULT
================================================================================
ID   Zone                     Request    Minimum    Priority   Allocated  Delivered  Shortage   Status
--------------------------------------------------------------------------------------------
Z01  Municipal Hospital       4000       3000       40.00      4082       4000       0          FULL
Z04  Ward 5 Residential Area  3800       1800       35.00      4318       3800       0          FULL
Z02  Central Flood Shelter    3500       2500       35.00      3646       3500       0          FULL
Z05  School Emergency Shelter 2600       1600       35.00      2737       2600       0          FULL
Z06  Fire and Emergency Service 2000       1500       35.00      2020       2000       0          FULL
Z03  Ward 3 Residential Area  4500       2000       30.00      4891       4500       0          FULL
================================================================================

Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

Summary
------------------------------------------
Total Requested Water          : 20400.00 L
Total Allocated Water          : 21694.00 L
Total Effective Water Delivered: 20400.00 L
Total Unresolved Requirement   : 0.00 L
Remaining Water                : 306.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 6 zone(s)
Minimum Requirement Satisfied: 0 zone(s)
Below Minimum                : 0 zone(s)
Completely Unserved          : 0 zone(s)
```

```
Check: total allocated (21694.00 L) does not exceed available supply (22000.00 L). OK.

Zone with the highest unresolved requirement: Z01 (Municipal Hospital) - shortage 0.00 L
```

Result: PASS — all 6 zones reached FULL, total effective delivered equalled total requested (20,400 L), and 306 L remained at the centre.

7.3 Case 3 — Supply insufficient even for the combined minimum

Expected: with only 10,000 L available against a 12,400 L combined minimum, at least one low-priority zone should fall BELOW MIN or UNSERVED, while the programme must still never over-allocate.

```
========== CASE 3 ==========
Insufficient Even For Combined Minimum Requirement
Available Water: 10,000 L

================================================================================
                                ALLOCATION RESULT
================================================================================
ID   Zone                     Request   Minimum   Priority  Allocated  Delivered  Shortage  Status
--------------------------------------------------------------------------------------------
Z01  Municipal Hospital       4000      3000      40.00     3061       3000       1000      MINIMUM
Z04  Ward 5 Residential Area  3800      1800      35.00     2045       1800       2000      MINIMUM
Z02  Central Flood Shelter    3500      2500      35.00     2604       2500       1000      MINIMUM
Z05  School Emergency Shelter 2600      1600      35.00     1684       1600       1000      MINIMUM
Z06  Fire and Emergency Service 2000      1500      35.00     605        599        1401      BELOW MIN
Z03  Ward 3 Residential Area  4500      2000      30.00     0          0          4500      UNSERVED
================================================================================


Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

Summary
------------------------------------------
Total Requested Water          : 20400.00 L
Total Allocated Water          : 10000.00 L
Total Effective Water Delivered: 9498.89 L
Total Unresolved Requirement   : 10901.11 L
Remaining Water                : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 4 zone(s)
Below Minimum                : 1 zone(s)
Completely Unserved          : 1 zone(s)

Check: total allocated (10000.00 L) does not exceed available supply (10000.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 4500.00 L
```

Result: PASS — 4 zones reached MINIMUM, Z06 fell to BELOW MIN, and Z03 (lowest priority) was completely UNSERVED; total allocated was exactly 10,000 L.

7.4 Case 4 — Two zones with an exact priority tie

Expected: “Shelter Alpha” and “Shelter Beta” are given identical type, requested, minimum and waiting-cycle values, forcing an exact tie in priority score. The tie-break rule should place Alpha immediately before Beta (input order), with both otherwise treated identically.

```
========== CASE 4 ==========
Equal Priority Tie-Break Test (Shelter Alpha vs Shelter Beta)
Available Water: 13,500 L


================================================================================
                                ALLOCATION RESULT
================================================================================
ID   Zone                     Request   Minimum   Priority  Allocated  Delivered  Shortage  Status
---------------------------------------------------------------------------------------------
ZA   Shelter Alpha            3000      2000      40.00     3158       3000       0         FULL
ZB   Shelter Beta             3000      2000      40.00     3158       3000       0         FULL
Z01  Municipal Hospital       4000      3000      40.00     4082       4000       0         FULL
Z06  Fire and Emergency Service 2000      1500      35.00     2020       2000       0         FULL
================================================================================


Priority Order:
1. ZA - Shelter Alpha (Priority 40.00)
2. ZB - Shelter Beta (Priority 40.00)
3. Z01 - Municipal Hospital (Priority 40.00)
4. Z06 - Fire and Emergency Service (Priority 35.00)


Summary
------------------------------------------
Total Requested Water          : 12000.00 L
Total Allocated Water          : 12417.62 L
Total Effective Water Delivered: 12000.00 L
Total Unresolved Requirement   : 0.00 L
Remaining Water                : 1082.38 L


Service Conditions
------------------------------------------
Full Requirement Satisfied   : 4 zone(s)
Minimum Requirement Satisfied: 0 zone(s)
Below Minimum                : 0 zone(s)
Completely Unserved          : 0 zone(s)


Check: total allocated (12417.62 L) does not exceed available supply (13500.00 L). OK.


Zone with the highest unresolved requirement: ZA (Shelter Alpha) - shortage 0.00 L
```

Result: PASS — both shelters share priority score 40.00, Alpha is ranked immediately above Beta, and both receive identical allocation, delivery and status values.

7.5 Case 5 — Increased distribution loss on one residential zone

Expected: raising Z04's loss from 12% to 60% means far more water must be released to deliver the same 1,800 L minimum, leaving less supply for the lowest-priority zone.

```
========== CASE 5 ==========
Increased Distribution Loss - Ward 5 Residential Area (Z04) Loss 12% -> 60%
Available Water: 13,500 L


================================================================================
                                ALLOCATION RESULT
================================================================================
ID   Zone                     Request   Minimum   Priority  Allocated  Delivered  Shortage  Status
---------------------------------------------------------------------------------------------
Z01  Municipal Hospital       4000      3000      40.00     3061       3000       1000      MINIMUM
Z04  Ward 5 Residential Area  3800      1800      35.00     4500       1800       2000      MINIMUM
Z02  Central Flood Shelter    3500      2500      35.00     2604       2500       1000      MINIMUM
Z05  School Emergency Shelter 2600      1600      35.00     1684       1600       1000      MINIMUM
Z06  Fire and Emergency Service 2000      1500      35.00     1515       1500       500       MINIMUM
```

```
Z03  Ward 3 Residential Area  4500      2000      30.00     135       124       4376      BELOW MIN
================================================================================

Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

Summary
------------------------------------------
Total Requested Water          : 20400.00 L
Total Allocated Water          : 13500.00 L
Total Effective Water Delivered: 10524.43 L
Total Unresolved Requirement   : 9875.57 L
Remaining Water                : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 5 zone(s)
Below Minimum                : 1 zone(s)
Completely Unserved          : 0 zone(s)

Check: total allocated (13500.00 L) does not exceed available supply (13500.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 4375.57 L
```

Result: PASS — Z04's allocation rose sharply (to 4,500 L released for the same 1,800 L delivered) and Z03, the lowest-priority zone, dropped from MINIMUM to BELOW MIN as a direct consequence.

7.6 Case 6 — Increased waiting period on a previously low-priority zone

Expected: raising Z03's previous waiting cycles from 2 to 8 should raise its priority score enough to overtake every other zone.

```
========== CASE 6 ==========
Increased Waiting Period - Ward 3 Residential Area (Z03) Waiting 2 -> 8
Available Water: 13,500 L

================================================================================
                                ALLOCATION RESULT
================================================================================
ID   Zone                     Request   Minimum   Priority  Allocated  Delivered  Shortage  Status
-------------------------------------------------------------------------------------------
Z03  Ward 3 Residential Area  4500      2000      60.00     2590       2383       2117      MINIMUM
Z01  Municipal Hospital       4000      3000      40.00     3061       3000       1000      MINIMUM
Z04  Ward 5 Residential Area  3800      1800      35.00     2045       1800       2000      MINIMUM
Z02  Central Flood Shelter    3500      2500      35.00     2604       2500       1000      MINIMUM
Z05  School Emergency Shelter 2600      1600      35.00     1684       1600       1000      MINIMUM
Z06  Fire and Emergency Service 2000      1500      35.00     1515       1500       500       MINIMUM
================================================================================

Priority Order:
1. Z03 - Ward 3 Residential Area (Priority 60.00)
2. Z01 - Municipal Hospital (Priority 40.00)
3. Z04 - Ward 5 Residential Area (Priority 35.00)
4. Z02 - Central Flood Shelter (Priority 35.00)
5. Z05 - School Emergency Shelter (Priority 35.00)
6. Z06 - Fire and Emergency Service (Priority 35.00)

Summary
------------------------------------------
Total Requested Water          : 20400.00 L
Total Allocated Water          : 13500.00 L
```

```
Total Effective Water Delivered: 12782.61 L
Total Unresolved Requirement   : 7617.39 L
Remaining Water                : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 6 zone(s)
Below Minimum                : 0 zone(s)
Completely Unserved          : 0 zone(s)

Check: total allocated (13500.00 L) does not exceed available supply (13500.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 2117.39 L
```

Result: PASS — Z03's priority score rose to 60.00 (20 criticality + 8×5 waiting bonus) and it moved from priority rank 6 to priority rank 1, becoming the first zone served.

7.7 Case 7 (student-designed) — Supply exactly equal to the raw minimum total

Expected: available water is set to exactly 12,400 L, numerically equal to the sum of the six raw minimum requirements. Because guaranteeing a minimum requires releasing minimum ÷ (1 − loss), the true loss-adjusted requirement is higher than this raw sum, so the lowest-priority zone should still fall short of its minimum — demonstrating that distribution loss silently reduces the effective capacity of a supply that looks sufficient on paper.

```
========== CASE 7 (Student-Designed) ==========
Available Water Exactly Equals Combined Raw Minimum Requirement (12,400 L)
Available Water: 12,400 L

================================================================================
                                ALLOCATION RESULT
================================================================================
ID   Zone                     Request   Minimum   Priority  Allocated  Delivered  Shortage  Status
--------------------------------------------------------------------------------------------
Z01  Municipal Hospital       4000      3000      40.00     3061       3000       1000      MINIMUM
Z04  Ward 5 Residential Area  3800      1800      35.00     2045       1800       2000      MINIMUM
Z02  Central Flood Shelter    3500      2500      35.00     2604       2500       1000      MINIMUM
Z05  School Emergency Shelter 2600      1600      35.00     1684       1600       1000      MINIMUM
Z06  Fire and Emergency Service 2000      1500      35.00     1515       1500       500       MINIMUM
Z03  Ward 3 Residential Area  4500      2000      30.00     1490       1371       3129      BELOW MIN
================================================================================

Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

Summary
------------------------------------------
Total Requested Water          : 20400.00 L
Total Allocated Water          : 12400.00 L
Total Effective Water Delivered: 11770.61 L
Total Unresolved Requirement   : 8629.39 L
Remaining Water                : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 5 zone(s)
Below Minimum                : 1 zone(s)
Completely Unserved          : 0 zone(s)

Check: total allocated (12400.00 L) does not exceed available supply (12400.00 L). OK.
```

```
Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 3129.39 L
```

Result: PASS — 5 zones reached MINIMUM while Z03 (priority rank 6) fell to BELOW MIN, confirming that raw totals can be misleading once distribution loss is taken into account.

8. Limitations

- The priority weights (40/35/30/20 for zone type, 5 points per waiting cycle) were chosen by reasoned judgement rather than derived from a formal decision-making model; a real municipality might calibrate these differently based on policy or medical guidance.
- The maximum number of zones is fixed at compile time (MAX_ZONES = 50), which is more than sufficient for a single distribution centre but would need to be changed (or replaced with dynamic memory allocation) for a much larger network.
- The model assumes distribution loss is a fixed percentage known in advance for each zone; in reality, loss could vary between delivery cycles due to weather, road conditions or equipment failure.
- The programme performs a single allocation pass for one supply of water; it does not model multiple distribution cycles over time or update previous waiting cycles automatically between runs.
- Input validation is limited to reading the values supplied; it assumes the person entering data provides non-negative, sensible numbers, as extensive input-sanitisation was outside the scope of the assignment's focus on data organisation and allocation logic.

9. Conclusion

The developed programme allocates a limited emergency water supply across competing service zones by combining a transparent, justified priority score (zone criticality plus previous waiting) with a two-phase, loss-aware release procedure that guarantees minimum requirements before extending any zone toward its full request. Because every decision is driven entirely by the input data rather than by hard-coded values, the same code correctly handled all seven test scenarios, from full sufficiency to severe shortage, exact ties in priority, and changes in distribution loss or waiting history. The solution therefore achieves the balance the brief asked for between emergency need, fairness and efficient use of the available water, while remaining simple enough to verify and extend.

10. References

No external text, code or dataset was copied for this assignment; the allocation strategy, algorithm and C implementation were developed independently by the student based on the assignment brief. The following general references were consulted for the C language features used.

[1] B. W. Kernighan and D. M. Ritchie, The C Programming Language, 2nd ed. Englewood Cliffs, NJ, USA: Prentice Hall, 1988.

[2] cppreference.com, “C Standard Library headers <stdio.h>, <string.h>.” [Online]. Available: https://en.cppreference.com/w/c/header